

Introduction to Stratification

Learn about stratification for causal inference.

Congratulations!

You have just been hired by the local health officer to estimate the effect of having high cholesterol on blood pressure. You have a dataset from a hospital and are eager to apply your new knowledge of causal inference.

The dataset includes the following variables on 100 patients:

A primer on blood pressure:

Your first task is to estimate the effect of having high cholesterol on systolic blood pressure.

Cholesterol is the treatment variable. Blood pressure is the outcome variable. Age and smoking status are *covariates* (variables that are not the variables of interest).

There are many risk factors for high blood pressure, including obesity, diabetes, age, and family history. This is a problem because you will have trouble studying cholesterol (the treatment) independently from these other factors.

Because we cannot randomize who has high cholesterol, there will likely be imbalances in other variables, between patients with high cholesterol (treatment group) and patients who do not have high cholesterol (control group). For example, overall the high cholesterol group might be older or more people might be smokers.

But you have some tools to address this problem. The first is Randomization.

Randomization makes the estimation of causal effects easier. In randomized studies, the covariates in the treatment and control groups look similar to each other (i.e., there just as many smokers are in both groups), so treatment status is the only difference between them. By contrast, in non-randomized studies, the variables of the treatment and control groups may look very different (i.e., more smokers in the high cholesterol group).

You can balance the covariates with matching and weighting methods.

One matching method is to first separate the data into subgroups, or *strata* (*stratification*), based on some variable such as smoking.

First, start by dividing the group into the smokers and non-smokers. Looking at just the smokers, compare the incidence of high cholesterol and high blood pressure.

Now do the same for the non-smokers. This article dives more into the specifics, but that is the basic idea: divide the data into groups based on some variable and compare within that group.

Because the individuals *within the subgroups* are similar to one another, you have satisfied the *conditional exchangeability assumption* of causal inference: if treatment

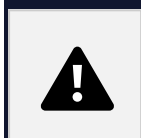
assignments were swapped, the outcomes would still be the same within each subgroup, on average.

Either categorical or numeric variables can be used to split the data into subgroups, but the process differs slightly depending on which type of variable is used.

Stratification on Categorical Variables

Nicotine (the chemical in tobacco products) causes physical changes that cause both high blood pressure and high cholesterol. Because smoking tobacco is related to both high blood pressure (outcome) and high cholesterol (treatment), you may not be able to estimate the effect of high cholesterol on blood pressure if you do not take smoking into account.

The following two plots show what stratifying the blood pressure data according to tobacco use looks like.



In the first plot, we see that the number of individuals with high cholesterol ($n = 52$) is approximately the same as the number who have normal cholesterol ($n = 48$).

However, the second plot shows that the ratio of smokers to non-smokers in each group is quite different. Among individuals with normal cholesterol, 33 (69%) are non-smokers; among those with high cholesterol, only 22 (42%) are non-smokers.

If we don't take smoking status into account when comparing the high and normal cholesterol groups, we would be comparing two very different groups. In other words, we would be violating the conditional exchangeability assumption of causal inference. This could lead to a biased estimate of the treatment effect because we would be ignoring the fact that smoking is related to blood pressure.

Estimation of Causal Effects

By separating the observations by smoking status, we have stratified our data into two subgroups. When using stratification, causal effects are not estimated directly from the entire sample, but rather within each subgroup. We can view this process as three steps:

The table below shows the calculation of the average blood pressure in the treatment and control groups for smokers and non-smokers.

Average Blood Pressure		
	Non-smoker	Smoker
High Cholesterol	139	155
Normal Cholesterol	110	117

Let's follow the three steps above to estimate the treatment effect of tobacco use when we stratify on smoking status.

First, we calculate the average treatment effect within each subgroup:

$$ATE_{Non-smokers} = 139 - 110 = 29 \text{ mmHg}$$

ATE

Non-smokers

$$= 139 - 110 = 29 \text{ mmHg}$$

$$ATE_{Smokers} = 155 - 117 = 38 \text{ mmHg}$$

ATE

Smokers

$$=155-117=38\text{mmHg}$$

Next, we multiply the subgroup ATEs by the proportions of each subgroup. In this case, out of 100 patients, 55 were non-smokers and 45 were smokers.

$$\text{Non-smokers } 29 \times 55 / 100 = 15.9 \text{ mmHg}$$

Non-smokers $29 \times$

100

55

$$=15.9\text{mmHg}$$

$$\text{Smokers } 38 \times 45 / 100 = 17.1 \text{ mmHg}$$

Smokers $38 \times$

100

45

$$=17.1\text{mmHg}$$

Finally, to estimate a causal effect for ALL patients, we sum the weighted subgroup ATEs.

$$\text{Weighted ATE} = 15.9 + 17.1 = 33 \text{ mmHg}$$

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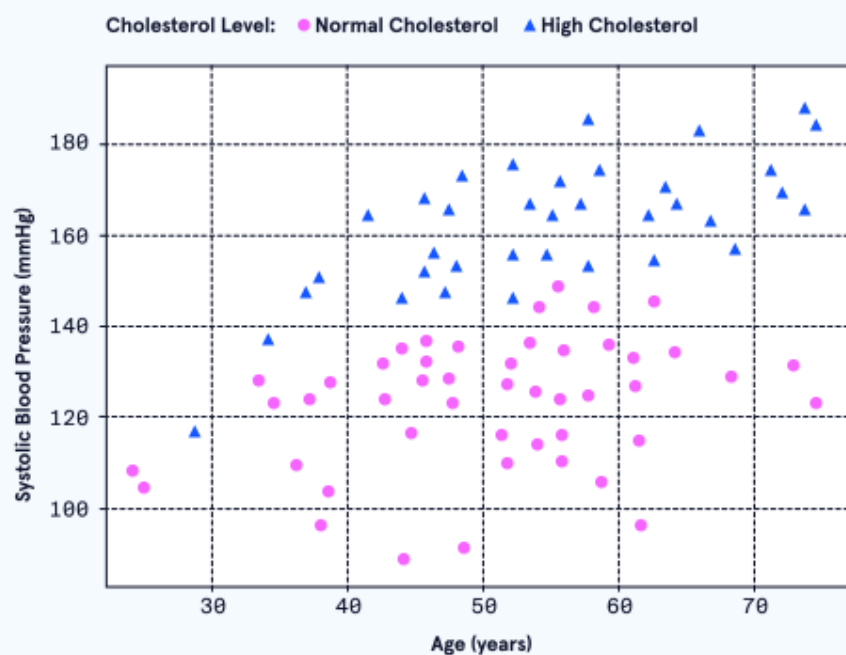
The weighted ATE of 33 mmHg suggests that, when accounting for smoking status, individuals with high cholesterol have on average a systolic blood pressure 33 mmHg higher than that of individuals with normal cholesterol.

Stratification on Numeric Variables

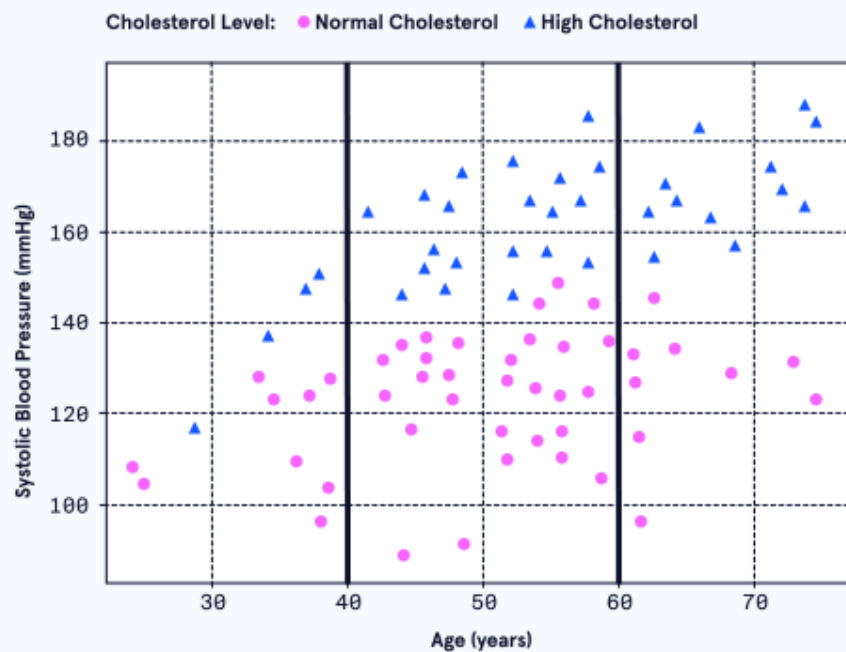
Stratification by numeric variables requires a little more effort. Unlike categorical variables, which have just a few subgroups, numeric variables can vary greatly. This means that without some kind of manipulation of the data, we could end up with a huge number of subgroups each containing very few observations.

Let's think about age first. If we have 100 people age 20-80 (a 60 year span), we will have 60 different ages to work with.

Rather than treating every unique age as its own subgroup, we could lump them together into ranges or *bins*. Let's start with three age groups: younger than 40, between 40 and 60, and older than 60.



Now that we have defined subgroups for age, we can estimate the average treatment effect of tobacco use. We can follow the same three steps we did when stratifying on smoking status.



We look at each subgroup separately and subtract the average blood pressure of the control group from the average blood pressure of the treatment group to get each subgroup's ATE.

$$\text{Age} < 40 \text{ ATE} = 138 - 116 = 22 \text{ mmHg}$$

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$$40 \leq \text{Age} \leq 60 \text{ ATE} = 154 - 121 = 33 \text{ mmHg}$$

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$$\text{Age} > 60 \text{ ATE} = 169 - 127 = 42 \text{ mmHg}$$

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Then, we weight each subgroup's ATE by the proportion of observations in that subgroup.

$$\text{Age} < 40 \text{ ATE} = 22 * (14/100) = 3.08 \text{ mmHg}$$

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$$40 \leq \text{Age} \leq 60 \text{ ATE} = 33 * (56/100) = 18.5 \text{ mmHg}$$

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$$\text{Age} > 60: 42 * (30/100) = 12.6 \text{ mmHg}$$

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Finally, we add the weighted averages together.

$$\text{Weighted ATE} = 3.08 + 18.5 + 12.6 = 34.1 \text{ mmHg}$$

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The weighted average treatment effect of 34.1 mmHg suggests that, when accounting for age, individuals with high cholesterol have on average a systolic blood pressure 34.1 mmHg higher than that of individuals with normal cholesterol.

Limitations

You may notice that we got different estimates for the ATE depending on which variable we used to stratify the data, which leads to an important question: which estimate should we trust?

The estimated ATE is just that — an estimate. We could refine our estimate of the ATE by stratifying on both smoking status AND age, which would require us to compute a weighted ATE estimate for each of the six strata. However, stratifying on too many variables comes with its own limitations.

Stratification is useful primarily in *low-dimensional* data, which is data where there are very few observed variables. Stratification on multiple variables can lead to very small subgroups, making estimation of causal effects less reliable. Further, stratification is not a particularly efficient use of data. If the overlap assumption is not met for a subgroup, we cannot estimate a causal effect using that data.

Assumptions

The three main assumptions for causal inference apply for stratification.

Review

In this article, we learned that: